

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: COMPUTER NETWORKS

COURSE CODE: CSA0714

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 30%

Annexure A

Scenario-Based

Code of Conduct:

I Renee Vincent ASR-192424161 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Renee Vincent ASR

Annexure A
Scenario-Based

1. A newly established Smart Campus is deploying a secure network infrastructure for its academic and administrative departments. The network administrator has been allocated the IP address block 192.168.100.0/24. Based on the expected number of devices, the following subnet masks have already been assigned:

School of Computing → /25

School of Electronics → /26

School of Mechanical Engineering → /27

As the network engineer, design the subnet allocation by performing the following tasks:

- 1. Determine the network address and broadcast address for each department.**
- 2. Identify the valid host IP address range for each subnet.**
- 3. Calculate the number of usable host addresses available in each subnet.**
- 4. Determine the remaining unused IP address range within the given network block.**

Given

- Network Block: 192.168.100.0/24**
- School of Computing: /25**
- School of Electronics: /26**
- School of Mechanical Engineering: /27**

1. Determine the Network and Broadcast Address

Department	Subnet	Network Address	Broadcast Address
School of Computing	192.168.100.0/25	192.168.100.0	192.168.100.127
School of Electronics	192.168.100.128/26	192.168.100.128	192.168.100.191
School of Mechanical Engineering	192.168.100.192/27	192.168.100.192	192.168.100.223

2. Valid Host IP Address Range

Department	Host IP Range
School of Computing	192.168.100.1 – 192.168.100.126
School of Electronics	192.168.100.129 – 192.168.100.190
School of Mechanical Engineering	192.168.100.193 – 192.168.100.222

3. Calculate the Number of Usable Host Addresses

Department	Subnet	Total Addresses	Usable Hosts
School of Computing	/25	128	126
School of Electronics	/26	64	62
School of Mechanical Engineering	/27	32	30

4. Remaining Unused IP Address Range

After allocating:

- **/25:** 192.168.100.0 – 192.168.100.127
- **/26:** 192.168.100.128 – 192.168.100.191
- **/27:** 192.168.100.192 – 192.168.100.223

Remaining addresses:

- **Network Range:** 192.168.100.224 – 192.168.100.255
- **Total Remaining Addresses:** 32 (equivalent to one /27 subnet)

2. A multinational software company is expanding its headquarters and has been assigned the private IP block 10.10.0.0/16. Different departments require different subnet sizes based on the number of employees. The network administrator has allocated the following subnet masks:

Software Development → /24

Quality Assurance → /25

Human Resources → /26

Executive Management → /27

As the network administrator, perform the following tasks:

- 1. Determine the network address and broadcast address for each department.**
- 2. Identify the valid range of host IP addresses.**

3. Calculate the number of usable hosts in each subnet.
4. Identify the unused IP address range remaining within the allocated IP block.

Given

- **Allocated IP Block:** 10.10.0.0/16
- **Software Development:** /24
- **Quality Assurance:** /25
- **Human Resources:** /26
- **Executive Management:** /27

1. Determine the Network Address and Broadcast Address

Department	Subnet	Network Address	Broadcast Address
Software Development	10.10.0.0/24	10.10.0.0	10.10.0.255
Quality Assurance	10.10.1.0/25	10.10.1.0	10.10.1.127
Human Resources	10.10.1.128/26	10.10.1.128	10.10.1.191
Executive Management	10.10.1.192/27	10.10.1.192	10.10.1.223

2. Valid Host IP Address Range

Department	Host IP Range
Software Development	10.10.0.1 – 10.10.0.254
Quality Assurance	10.10.1.1 – 10.10.1.126
Human Resources	10.10.1.129 – 10.10.1.190
Executive Management	10.10.1.193 – 10.10.1.222

3. Calculate the Number of Usable Hosts

Department	Subnet	Total Addresses	Usable Hosts
Software Development	/24	256	254
Quality Assurance	/25	128	126
Human Resources	/26	64	62
Executive Management	/27	32	30

4. Identify the Remaining Unused IP Address Range

Allocated ranges:

- 10.10.0.0 – 10.10.0.255 (/24)
- 10.10.1.0 – 10.10.1.127 (/25)
- 10.10.1.128 – 10.10.1.191 (/26)
- 10.10.1.192 – 10.10.1.223 (/27)

Unused addresses within the allocated portion:

- 10.10.1.224 – 10.10.1.255 (32 addresses)

Unused addresses remaining in the overall 10.10.0.0/16 block:

- 10.10.2.0 – 10.10.255.255

3.A smart hospital is migrating its network infrastructure from IPv4 to IPv6 to support thousands of connected medical devices and IoT sensors. The hospital has been assigned the IPv6 network:

2001:0db8:abcd:0000:0000:0000:0000/64

As the network engineer responsible for the migration, perform the following:

- 1. Convert the given IPv6 address into its compressed notation.**
- 2. Expand the compressed address back into its full hexadecimal representation.**
- 3. Identify the network prefix using the /64 prefix length.**
- 4. Calculate the total number of host addresses supported within the subnet.**
- 5. the total number of possible host addresses available within this network.**

Given

- **IPv6 Network:** 2001:0db8:abcd:0000:0000:0000:0000/64

1. Convert the IPv6 Address into Compressed Notation

Remove leading zeros in each hextet and replace consecutive zero hexkets with ::.

Compressed IPv6 Address:

2001:db8:abcd::/64

2. Expand the Compressed Address

Expand 2001:db8:abcd::/64 back to its full form.

Expanded IPv6 Address:

2001:0db8:abcd:0000:0000:0000:0000/64

3. Identify the Network Prefix

With a /64 prefix length:

- **Network Prefix: 2001:db8:abcd:0::/64**
- The first **64 bits** (first four hextets) represent the network.
- The remaining **64 bits** represent the host/interface ID.

4. Calculate the Number of Host Addresses Supported

Host bits = $128 - 64 = 64$

Number of host addresses:

$2^{64} = 18,446,744,073,709,551,616$

5. Total Number of Possible Host Addresses

The total number of interface (host) addresses available within this /64 IPv6 network is:

18,446,744,073,709,551,616 (2^{64})

4. An Internet Service Provider (ISP) connects multiple customer networks through a core router. The router maintains the following routing table entries:

192.168.1.0/24

192.168.2.0/24

192.168.3.0/24

A packet arrives at the router with the destination IP address:

192.168.2.45

As the routing engineer, determine:

- 1. The destination subnet to which the packet belongs.**
- 2. The matching routing table entry.**
- 3. The next-hop network/interface used to forward the packet.**
- 4. The default route that should be used if the destination does not match any routing table entry.**

Given

Routing Table

Destination Network
192.168.1.0/24
192.168.2.0/24
192.168.3.0/24

Destination IP Address: 192.168.2.45

1. Determine the Destination Subnet

The destination IP **192.168.2.45** falls within:

- **Network Address:** 192.168.2.0
- **Broadcast Address:** 192.168.2.255
- **Host Range:** 192.168.2.1 – 192.168.2.254

Destination Subnet: 192.168.2.0/24

2. Identify the Matching Routing Table Entry

Using the **longest-prefix match**, the router selects:

192.168.2.0/24

3. Determine the Next-Hop Network/Interface

Since the routing table lists only destination networks (no next-hop addresses are provided), the packet is forwarded through the **interface connected to the 192.168.2.0/24 network**.

Next-Hop Network/Interface: Interface connected to 192.168.2.0/24

4. Default Route

If the destination IP does **not** match any routing table entry, the router forwards the packet using the **default route**.

Default Route: 0.0.0.0/0 (Gateway of Last Resort)

5. A company establishes two office departments connected through a Layer-3 router. The allocated subnet information is as follows:

Administration LAN → 192.168.10.0/26

Finance LAN → 192.168.10.64/26

The router interfaces are configured with:

Administration Gateway → 192.168.10.1

Finance Gateway → 192.168.10.65

As the network administrator, perform the following tasks:

- 1. Determine the network address and broadcast address for each LAN.**
- 2. Identify the valid host IP address range.**
- 3. Suggest suitable IP addresses for three computers in each LAN.**
- 4. Identify the default gateway that hosts in each LAN should use for communication.**

Given

- Administration LAN: 192.168.10.0/26**
- Finance LAN: 192.168.10.64/26**

Router Interfaces

- Administration Gateway: 192.168.10.1**
- Finance Gateway: 192.168.10.65**

1. Determine the Network Address and Broadcast Address

LAN	Network Address	Broadcast Address
Administration LAN	192.168.10.0	192.168.10.63
Finance LAN	192.168.10.64	192.168.10.127

2. Identify the Valid Host IP Address Range

LAN	Valid Host Range
Administration LAN	192.168.10.1 – 192.168.10.62
Finance LAN	192.168.10.65 – 192.168.10.126

3. Suggest Suitable IP Addresses for Three Computers

Administration LAN

Computer	IP Address
PC-1	192.168.10.2
PC-2	192.168.10.3
PC-3	192.168.10.4

Finance LAN

Computer	IP Address
PC-1	192.168.10.66
PC-2	192.168.10.67
PC-3	192.168.10.68

4. Identify the Default Gateway

LAN	Default Gateway
Administration LAN	192.168.10.1
Finance LAN	192.168.10.65

RUBRICS FOR CALCULATION

Scenario Based Assessment					
Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding ; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand